

Stacking and Blending

⇒ Stacking: Here we train our base models on the dataset given to us. And so we have predicted output of each model we used. This updated dataset is given to the meta model for training.

Now, for prediction on any data the base model first gives prediction and send it to meta model for final prediction.

⇒ How stacking is diff. from Bagging or Boosting?

- It can have diff. base models
- The training is done twice here.

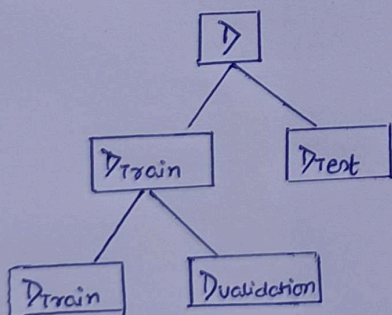
⇒ What's the problem with stacking? The output given to meta model be overfitted.

⇒ What's the solution? → called Blending

① Hold out method: a part of data is used to train base models and rest to train meta model.

② K-fold method → called stacking

⇒ Hold out Approach - Blending:



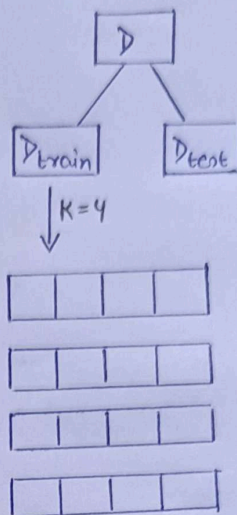
Step1: Train 3 (say) base model on D_{train}

Step2: Base model will predict on $D_{validation}$ and give new dataset

Step3: Train the meta model on this new data set, and while prediction we use D_{test} .

This whole method will reduce overfitting.

⇒ K-fold Approach - Stacking:



Step1: Train one of the base model on $K-1$ set and then predict it on the left even data. Again, we will train the model on $K-1$ set and this time we will change the left even dataset, and then we test on the new left even dataset. We do it on each K .

Now, we do same for next base model.

Step2: Now we train the meta model on this updated dataset.

Step3: We will train base model again with D_{train} , and predict on D_{test} .

Thus, the meta model is trained first and then base models are trained.

⇒ Multi-layer stacking:

